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A Novel GAN-Based Technique Towards Smarter Seismic Inversion

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Abstract

Seismic surveying is a crucial and effective method for exploring hydrocarbon reserves, continually attracting the attention of the upstream oil and gas sectors. Seismic inversion, a pivotal technique for investigating the properties of underlying geological strata, presents a significant problem. This work diverges from the traditional strategy of solely improving or amalgaming existing seismic inversion techniques.

We have utilized a novel generative-adversarial network (GAN) method, a deep learning technique, specifically trained for the seismic inversion process. This unique methodology seeks to eliminate certain critical obstacles, such as the calculation of the inversion matrix, the initialization of the wavelet, and the management of limitations associated with the restricted frequency band of seismic amplitudes in seismic inversion. This research has produced significant outcomes. By employing the generative-adversarial deep learning method, we have successfully addressed the previously described obstacles and attained remarkable quality and accuracy in our outcomes. We performed seismic inversion utilizing actual data from an oil field, attaining a remarkable accuracy rate of 97.5%. The accuracy percentage is corroborated by the validation data and the mean squared error (MSE), so strengthening the reliability of the suggested method. Moreover, the acoustic impedance of the five test wells consistently registered below 0.125 units, underscoring the superiority of our results. The correlation coefficient among these test wells varied from a minimum of 96% to a maximum of 99%. The acoustic impedances derived from the band-limited technique and the model-based method exhibited correlation coefficients of 71% and 83%, respectively. The application of the generative-adversarial algorithm in the inversion process highlights its smart modern effectiveness. It possesses the capacity to completely update and enhance the traditional seismic inversion method, heralding a new epoch in seismic exploration and inversion.

Keywords: Seismic Inversion, generative-adversarial network (GAN) algorithm, Deep Learning, Seismic Exploration, Seismic Reservoir characterization

Introduction

The pursuit of hydrocarbon reserves has significantly influenced the advancement of geophysical techniques, especially seismic methods (Russell, 1988; Chopra & Marfurt, 2005). Precisely imaging and describing subsurface structures is essential for identifying and evaluating hydrocarbon reserves (King, 1992; Mallick, 1995). The basis of seismic exploration was established by pioneering research in theoretical seismology. Researchers Ludger Mintrop and Sheriff conducted tests with artificial seismic sources, yielding significant insights into subsurface structures by employing explosives to generate seismic waves in 1949 (Mintrop, 1949; Sheriff & Gedart, 1995). The extensive utilization of reflection seismology accelerated in the mid-20th century owing to notable progress in digital recording technology and the simultaneous recognition of its substantial capacity for comprehensive imaging of subterranean geological formations. The seismic inversion method is complex and time-intensive due to the large amount of data needed for its calculations. As a result, many obstacles necessarily emerge during seismic inversion (Russell, 1988; Yilmaz, 2001). Concurrently, seismic inversion techniques began to develop in response to the need for quantitative subsurface data.

Initial inversion endeavors, exemplified by the restricted band inversion technique, aimed to extract fundamental characteristics such as acoustic impedance from seismic data (Ferguson & Margrave, 1996; Helgesen et al., 2000; Maurya & Singh, 2017). The ongoing study on seismic inversion revealed a variety of innovative methods, each characterized by distinct features. The base and sparse spike model, designed to depict subsurface characteristics as a sequence of spikes in depth, represented a notable shift from linearized approximations (Ji, Yuan, & Wang, 2019; Kushwaha et al., 2020). This method established the basis for contemporary seismic inversion techniques, integrating the complete wave equation and employing optimization methods. The late 20th and early 21st centuries had a convergence of increased computational capabilities, intricate algorithms, and superior data collecting devices. This collaboration resulted in the advancement of 3D and 4D seismic surveys, enabling the visualization of underlying structures with unparalleled precision. Seismic inversion progressed along with these advancements, with methodologies like full-waveform inversion (FWI) becoming increasingly significant (Virieux & Operto, 2009; Xu et al., 2012; Lin et al., 2023; Azizzadeh Mehmandost Olya et al., 2024a).

As previously stated, numerous techniques have been developed to assign this duty to computers. The essential issue is that these algorithms are incapable of learning or demonstrating intelligent behavior. As a result, their parameters require manual modification for each occurrence by an operator. Moreover, when encountering difficulties during the inversion process, these algorithms are unable of autonomously addressing problems. Moreover, it is crucial to recognize that specific traditional seismic inversion methods are vulnerable to noise, resulting in the dissemination of mistakes throughout the inversion process (Gogoi & Chatterjee, 2019; Zhang et al., 2023; Lin et al., 2023). Due to their exceptional capabilities, deep learning algorithms offer a highly promising approach for the application of intelligent techniques in seismic inversion. These advanced algorithms has the capacity to transform the entire seismic inversion process. Deep learning has gained significant appeal in recent years because to its extraordinary capabilities in the exploration of hydrocarbon deposits (Bagheri et al., 2024; Moradi Chaleshtori et al., 2024; Bahramali Asadi Kelishami & Mohebian, 2021; Mohebian et al., 2021). This advanced technology is transforming our methodology for locating and evaluating these precious natural resources. The increasing significance can be ascribed to multiple essential variables (Zhu, et al., 2022). Deep learning algorithms are proficient in analyzing extensive and intricate datasets, which is essential in hydrocarbon exploration. Deep learning significantly improves our ability to evaluate seismic data, geological formations, and subterranean conditions with unparalleled precision, thereby enabling us to identify possible reserves and assess their feasibility (Azizzadeh Mehmandost Olya, et al., 2024b; Rajabi, et al., 2019). Furthermore, the versatility of deep learning models renders them essential for addressing the intrinsic issues of hydrocarbon exploration. These algorithms can perpetually evolve and enhance their prediction powers by assimilating fresh data, rendering them exceptionally effective in adapting to changing geological circumstances and exploration methodologies (Geng & Wang, 2020; Habibullah, et al., 2020). In the domain of hydrocarbon reserve exploration (Azizzadeh Mehmandost Olya & Mohebian, 2023a) and the complex analysis of seismic data, deep learning algorithms have achieved significant application, highlighting the unique importance of this category of algorithms (Azizzadeh Mehmandost Olya & Mohebian, 2023b).

This paper utilized the Cyclic Generative Adversarial Network (GAN) deep learning technique. This algorithm exemplifies a state-of-the-art methodology, characterized by its distinctive functionalities. Its origin is traced to 2014, attributed to Ian Goodfellow (Goodfellow, et al., 2014). The algorithm currently features multiple unique architectures, indicating its continuous progress. The principal motivation for developing this algorithm has been the production of realistic photographs. This program may create previously nonexistent images after enduring extensive training and exposure to various visual inputs (Frank, et al., 2020).

In 2019, Xu and his colleagues successfully converted this technique for tabular data, demonstrating its adaptability beyond image data (Xu et al., 2019). In 2020, Liu and his research team accomplished a significant achievement by utilizing the GAN algorithm to develop a model for weather estimate and forecasting. This achievement highlights the algorithm's remarkable computational capabilities (Liu & Lee, 2020). This study employs the GAN algorithm to generate acoustic and density logs, so enabling seismic inversion through an innovative method. This strategy not only avoids the usual obstacles associated with traditional methods but also improves accuracy and considerably accelerates the process.

Methodology

The GAN is an advanced deep-learning method that provides multiple variations designed for certain applications. This paper utilizes the cyclic variation and will clarify its design (Creswell et al., 2018; Aggarwal et al., 2021). The GAN method fundamentally consists of two separate sub-networks that function simultaneously. The interaction between the two sub-networks enhances the accurate training of the intended models and commands (Gui et al., 2021). An apt analogy to clarify this algorithm is that of a forger and an investigator. In an imagined mental domain, the forger seeks to produce the most convincing counterfeit currency, while the investigator aims to differentiate counterfeit money from authentic currency. The detective notifies the counterfeiter via a communication channel of the identification of the counterfeit currency. The detection of counterfeit money signifies that the counterfeiter has to enhance their skills, while the detective is successfully executing their duties. If the detective fails to correctly identify the counterfeit currency, it indicates a necessity for further training. The ideal condition is reached when the counterfeiter is unable to improve the resemblance of their counterfeit currency to authentic money any further, and the investigator cannot distinguish the counterfeit from legitimate currency due to their closely aligned attributes. The example demonstrates two fundamental sub-networks: the generative and the adversarial.

The generative network improves result accuracy by producing requested samples and sending them to the adversarial network, which attempts to detect mistakes in the received examples. Moreover, within the context of this generative network technique, the network endeavors to comprehend the fundamental notions of the designated subject. This dynamic indicates that the network has the ability to produce genuine data that did not previously exist (Karras, et al., 2020). Figure 1 illustrates a simplified flowchart of the GAN algorithm. This flowchart depicts the previously discussed relationship between the generative subnet and the adversarial subnet. This graphic illustrates that when the data produced by the producing sub-network contradicts reality, the adversarial network requests the data to be regenerated.

The variety of algorithms in GAN sub-networks is contingent upon their objectives, and the selection of algorithms is essential for efficient learning. This paper employed fully connected neural networks for both the generative and adversarial sub-networks. The generator network employed the Rectified Linear Unit (ReLU) activation function to highlight nonlinear interactions. The many algorithms in GAN sub-networks facilitate a customized approach, allowing for task-specific adaptations. The implementation of fully linked neural networks in both sub-networks facilitates extensive information flow within the model's architecture. Incorporating the ReLU activation function into the generator network improves the model's capacity to discern intricate data patterns. The deliberate choice of algorithms demonstrates the precise design of GANs, where the interaction between components greatly affects the system's performance. Prior to examining the loss functions that regulate these subnetworks, it is imperative to grasp the notion of Nash equilibrium, a fundamental principle for understanding convergence in responses.

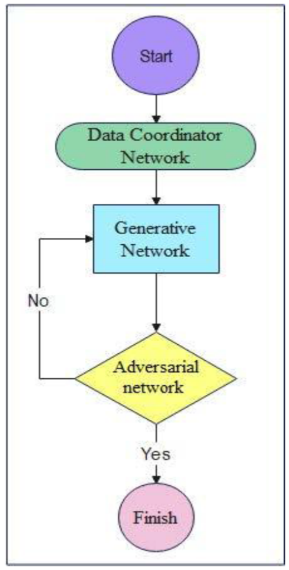

Figure 1—Streamlined flowchart of the GAN algorithm

Nash Equilibrium

Nash Equilibrium, a foundational concept in game theory, is a state in which each participant's strategy is optimal, given the strategies chosen by all other participants. Mathematically, in a strategic interaction involving multiple players, let's denote the strategy of player (i) as S_i, and the set of strategies of all players except (i) as S_{-i} The payoff to player (i) for playing strategy S_i while others play strategies from S_{-i} is denoted as $U_i\ (S_i, S_{-i})$. A Nash Equilibrium is reached when no player has an incentive to unilaterally change their strategy, i.e., for all players (i) the following condition holds:

$$U_i(S_i, S_{-i}) > U_i(\acute{S}_i, S_{-i})$$

for all possible alternative strategies $\acute{S}_i$ of player (i) and all combinations of strategies $\acute{S}_i$ of other players. In simple terms, each player's strategy is the best response to the strategies of others.

The concept of Nash Equilibrium appears intriguingly within the setting of Generative Adversarial Networks (GANs). Generative Adversarial Networks (GANs) comprise two fundamental subnetworks: the generator and the discriminator. The generator produces data instances, whereas the discriminator (Adversarial) assesses whether a certain instance is authentic (originating from the actual data distribution) or counterfeit (produced by the generator). This interaction resembles a competition between the generator and the discriminator. The generator endeavors to produce data indistinguishable from authentic data, whereas the discriminator strives to accurately discern between actual and counterfeit data. The discriminator's strategy mathematically involves establishing an optimal decision boundary, whereas the generator's strategy consists of generating data that can successfully mislead the discriminator. The equilibrium in this context refers to the condition when the generator provides data that is sufficiently realistic for the discriminator to be unable to differentiate it from authentic data, resulting in the discriminator's classification accuracy hovering around 50%, as it lacks the confidence to distinguish between the two. Attaining Nash Equilibrium in Generative Adversarial Networks is a complex endeavor. If the generator becomes excessively proficient, the discriminator may struggle to enhance its accuracy, and vice versa. This tug-of-war frequently results in fluctuations during training and difficulties in convergence. Researchers have devised multiple techniques to stabilize training, such as implementing mini-max game formulations and modifying learning rates.

Loss function

In the proposed GAN framework for seismic inversion, two neural networks the Generator G and the Discriminator D are trained simultaneously in an adversarial setting. The generator aims to produce synthetic density and sonic logs that are indistinguishable from real well-log data, while the discriminator strives to accurately classify inputs as real or generated.

The discriminator loss is defined as:

$$L_D = -E_{x \sim Preal}[\log D(x)] - E_{z \sim p_z}[\log(1 - D(G(Z)))]$$

where x denotes real data samples, z is noise input, $D(x)$ is the probability of a sample being real, and $G(z)$ is the generator's output. This formulation maximizes the likelihood of correctly identifying real data and minimizes the likelihood of incorrectly accepting generated data.

Conversely, the generator seeks to minimize:

$$L_G = E_{z \sim p_Z}[\log(D(G(Z)))]$$

forcing G to generate outputs that the discriminator classifies as real, i.e., $D(G(Z)) \rightarrow 1$ Training both networks is expressed as the minimax optimization problem:

$$min_G max_D V(D, G) = -E_{x \sim Preal}[\log D(x)] + E_{z \sim p_z}[\log(1 - D(G(Z)))]$$

Equilibrium is reached when neither network can improve unilaterally a state analogous to Nash equilibrium with D operating at ~50% accuracy because G generates samples closely matching the real data distribution. In the seismic inversion application, the generator operates in two stages. In the first cycle, high- and low-frequency acoustic impedance are derived separately from the nearest seismic trace T and well logs W:

$$AI_{HF,LF} = P_{inv}(T, W)$$

Where P_{inv} is the local inversion operator. Separating frequency components prevents low-frequency information from being overshadowed during learning. In the second cycle, the inversion is reconstructed from seismic traces using forward modeling:

$$\widehat{W} = P_{fwd}(T, AI)$$

Where P_{fwd} denotes the forward modeling process. The adversarial loss in this context incorporates both near-well and away-from-well seismic data:

$$L_{adv} = -E_{\acute{T} \notin near-well}[\log D(\acute{T})] - E_{T_{near-well}}[\log 1 - D\big(G(T)\big)]$$

This design ensures that the generator learns realistic log structures applicable across the seismic volume while the discriminator enforces data fidelity. Additionally, to prevent overfitting, neuron dropout is applied dynamically so that not all neurons converge to an optimal state within a single epoch, improving generalization and stability of convergence.

Seismic inversion process with GAN algorithm

Seismic inversion is a principal approach for investigating hydrocarbon reserves and underground water sources. Nevertheless, traditional techniques such as restricted band and model-based methods encounter a range of intrinsic limitations. These issues include wavelet extraction, limitations on data bandwidth, complex matrix computations (ill-conditioned matrices), and the pervasive problem of data noise.

This research presents an innovative method for seismic inversion designed to overcome these challenges. The selected methodology utilizes the GAN algorithm. Unlike previous methods that aimed to derive acoustic impedance from seismic traces, our approach entails the direct generation of acoustic impedance from these traces. At each site with a seismic trace, we acquire two logs: a density log and a sonic log. This new strategy obviates the necessity for wavelet production through several statistical methods. Furthermore, it eliminates the dependence on matrix inversion for seismic inversion, thus avoiding the related complications of matrix inversion. Moreover, the distinctive features of the GAN algorithm provide us with the additional benefit of reducing noise impacts during the inversion phase. This approach signifies a significant advancement in seismic inversion, tackling critical obstacles and improving the accuracy of hydrocarbon resource and subterranean water source discovery. Figure 2 presents a meticulously detailed flowchart, serving as a visual depiction of the core methodologies and important processes that drive the seismic inversion discussed in this paper. This visual exposition clarifies the incorporation of the GAN algorithm into the seismic inversion framework, providing users with a comprehensive knowledge of the complex stages involved in this novel methodology. In the following section, the goal is to deliver a thorough explanation of the unique components that constitute this complex flowchart. We intend to provide a clear and comprehensive knowledge of the visual representation through a deep analysis of each component.

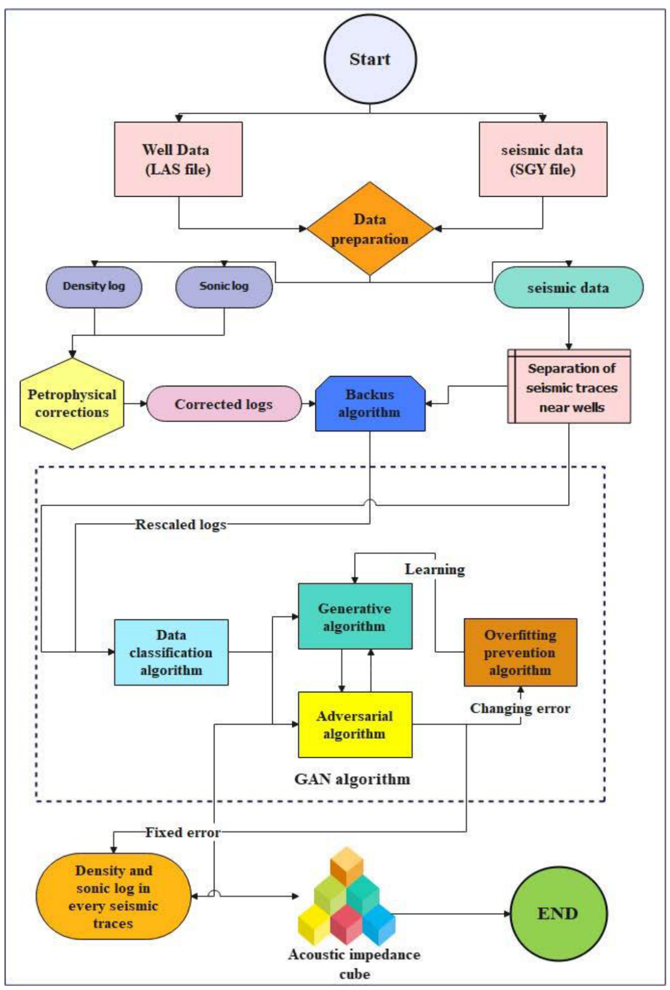

Figure 2—An illustration of the GAN algorithm's seismic inversion flowchart

Data preparation

This approach utilizes two principal data categories: seismic data and well log data. Python is our selected programming language because of its open-source characteristics and its ability to manage intricate deep-learning algorithms.

In the preliminary phases of this study, acquiring seismic and well data is essential. Due to the substantial amount of seismic data, we employ a data batching technique. This technology fulfills two functions: it diminishes the volume of data packets transmitted to the graphics processing unit and the main code kernel, therefore averting possible memory overflow in the computer. A vital factor is the distinct formatting of seismic data files (SGY format) and well-log data files (LAS format). These files are subjected to an advanced encryption procedure that considerably diminishes the storage demands for seismic and well-log data. Consequently, traditional methods for accessing these data are inadequate. To surmount this encryption obstacle and get the essential seismic and well-logging datasets, we devised a bespoke code customized for the specific encryption techniques and attributes of SGY and LAS files.

Backus algorithm for upscaling

After the thorough rectification of petrophysical log data and the careful separation from the nearest trace to the well data, we face one of the most critical hurdles in our pursuit of data alignment and integration. It is crucial to acknowledge that well logging instruments meticulously document data points at intervals of 0.1524 meters, while seismic sampling acquires data at somewhat broader intervals. Therefore, integrating these two distinct data sources requires a coordinated effort. Prior to commencing this undertaking, it is essential to solve the underlying issue of scale disparity between well logging and seismic data. Wellbore data is inherently measured in meters, while seismic data is inherently measured in milliseconds. To bridge this gap and create a unified scale, we utilize the essential resource of check shot data. Check shot data is essential for synchronizing these different scales, enabling accurate conversion between the meter and millisecond domains. Our primary purpose, which is to execute a seismic data inversion, depends on the flawless alignment attained through this technique. To get well data that corresponds with the seismic scale, we utilize a precisely formulated equation, rigorously developed using curve fitting to the check shot data (as illustrated in Figure 3). This formula acts as a conduit that guarantees our seismic data inversion endeavors are supported by precise and coordinated borehole data.

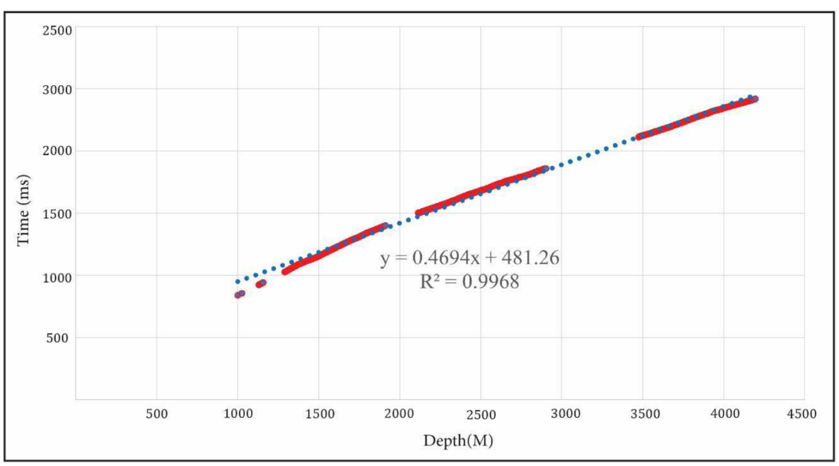

Figure 3—Curve fitting on check shot data

After successfully employing check shot data to convert borehole measurements from "meters" to "milliseconds," our next objective is to create a seamless link between this changed borehole data and the seismic dataset. To do this, we employ a clever technique proposed by George Backus in 1962, referred to as the Backus mean (Backus, 1962). The Backus mean approach offers a robust framework for connecting well log data with seismic data. It provides a systematic methodology for meaningfully relating and analyzing these two dissimilar datasets. This technique acts as a crucial link that integrates well log data with seismic data, hence improving our capacity to read and invert seismic data (Figure 4).

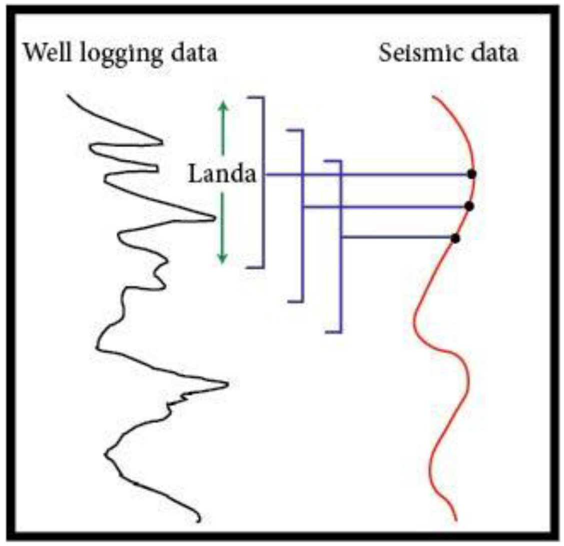

Figure 4—Backus mean

In the context of the Backus method, a vital parameter referred to as the λ value delineates the temporal scope of the window during which well data is closely integrated and correlated with seismic data. The length of this window is fundamentally linked to the temporal attributes of the collected seismic data. For example, if there is a *2.5-millisecond* pause between each recorded seismic domain, the associated window length will exactly equate to this *2.5-millisecond* duration. The precise alignment of the seismic data recording intervals with the Landa value supports the algorithm's efficacy. Furthermore, the transformation of well-logging data into the "milliseconds" scale enables a seamless integration within this timeframe. This harmonization guarantees the window functions flawlessly, facilitating the exact alignment and correlation of well data with its seismic equivalent.

Data classification algorithm

This research endeavor includes a data set of 15 wells and one onshore seismic cube. For our research and evaluation, we isolated 5 wells from the dataset, chosen for their unique dispersion properties, designating them for our testing phase. The remaining ten wells were allocated as training data. The proper training of the algorithm requires the precise classification and organization of these data points alongside the data from the nearest seismic trace. This methodology guarantees that, during the algorithm's training phase, each data point is meticulously aligned with analogous data, so augmenting the learning process. To accomplish this objective, we utilized a sophisticated data sorting method alongside the nearest seismic trace data. This rigorous sorting procedure underpins the development of a training feed for both the generating sub-networks and the adversarial sub-network. Organizing the data in this manner enhances the algorithm's ability to identify patterns, correlations, and subtleties, resulting in more robust and accurate outcomes in our exploration and analysis.

GAN algorithmic structure for seismic inversion

The GAN technique designed for seismic inversion is an advanced method utilizing a fully connected neural network. The generative subnet comprises three layers, each containing 1500 neurons. This particular configuration was meticulously selected to address the inherent complexity of seismic data, undergoing rigorous trial and error to adequately capture the numerous patterns and nuances within the data. The selection of the Rectified Linear Unit (ReLU) activation function for the generative method in this research is very significant.

This selection was driven by its proven capacity to harmoniously correspond with seismic data attributes, as revealed by the findings of Wang et al. (Wang et al., 2022). This activation function is crucial in determining the network's capacity to extract significant elements from the data, hence improving the quality and precision of the generated outcomes.

To comprehend the fundamental mechanics of the GAN algorithm, we will delineate its principal stages. The algorithm commences with "Input Gathering," during which it assembles essential input parameters and data. This phase establishes the foundation. The subsequent phase is "Generator Model Creation," during which the algorithm develops the generative model, considering the collected input. This influences the algorithm's functionalities.

Subsequently, the algorithm proceeds to "Layer Specification," wherein it delineates the network's architecture by defining layer configurations and dimensions. These specifications are essential for the model's optimal performance. Subsequently, we go to "Layer Creation," wherein the distinct levels of the network are instantiated in accordance with the specified design. The method emphasizes "Input Integration," seamlessly combining inputs from several layers to enhance data flow. It subsequently "Adds Connected Layers," promoting synergy among layers for collaborative processing of input data. The selected activation function, ReLU, incorporates non-linearity, augmenting the network's capacity to discern intricate patterns within the data.

The quantity of layers does not adhere to a predetermined pattern; it employs "Iterative Layer Addition," facilitating flexibility to data complexities. The algorithm attains the "Output Layer Definition" to delineate the output layer, consistent with research aims. The addition of this layer completes the network's architecture, guaranteeing that the model's output corresponds with the study objectives.

The outcome of these procedures is the "Final Generative Model," which includes both input and output layers. The approach culminates in the "Returning the Generator Model," providing a fully equipped generative model, prepared to produce valuable seismic inversion outcomes.

The ReLU activation function, recognized for its simplicity and efficacy, introduces non-linearity into neural networks. It effectively represents complex data relationships, rendering it appropriate for applications such as seismic inversion, where it excels in capturing detailed data patterns. ReLU is a non-linear activation function that incorporates non-linearity into the neural network. In contrast to other activation functions such as the sigmoid or hyperbolic tangent (tanh), ReLU possesses a straightforward mathematical formulation: it returns the input value if positive and zero if negative (Figure 5).

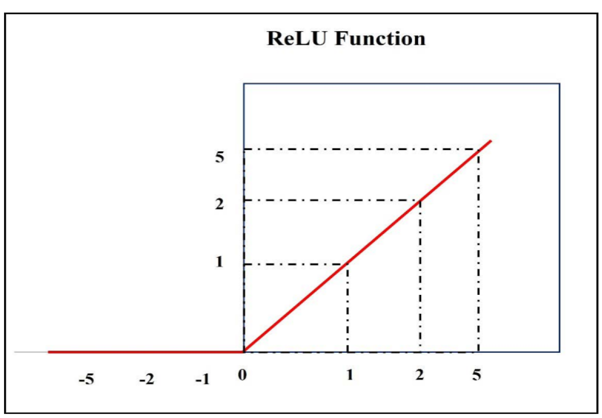

Figure 5—ReLU activation function

Result and discussion

We use a total of 15 wells, of which 10 were allocated for the training of the GAN algorithm. The remaining five served as test data repositories, crucial for assessing the algorithm's performance. It is essential to emphasize that each training well, along with its nearest seismic trace, provided 80% of its data for the training process, while the remaining 20% was crucial for the validation phase. This validation approach was effortlessly incorporated into the algorithm's continuous training protocol.

The convergence validation values produced by the algorithm offer a distinct and quantitative representation of its precision. These values not only function as a gauge of accuracy but also operate as an inherent mechanism for concluding the training process. The algorithm's training persists until the validation loss function values demonstrate no further substantial variations. When the algorithm arrives at a point where it can no longer minimize the loss function, the training process terminates smoothly.

In the context of our research, the algorithm attained a significant validation of the loss function following an extensive 3873 training iterations, resulting in a minimal loss value of 0.025. This achievement highlights the algorithm's remarkable accuracy, yielding an excellent score of 0.975.

Refer to Figures 6 and 7 for a visual representation of the algorithm's dynamic performance during each training phase. These charts clearly depict the variations in the loss function and the algorithm's accuracy across its training iterations.

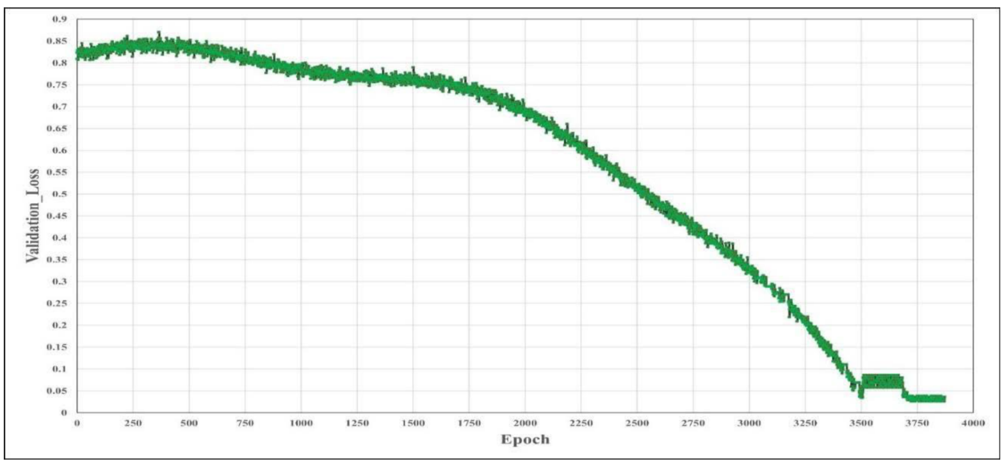

Figure 6—Validation loss per Epoch

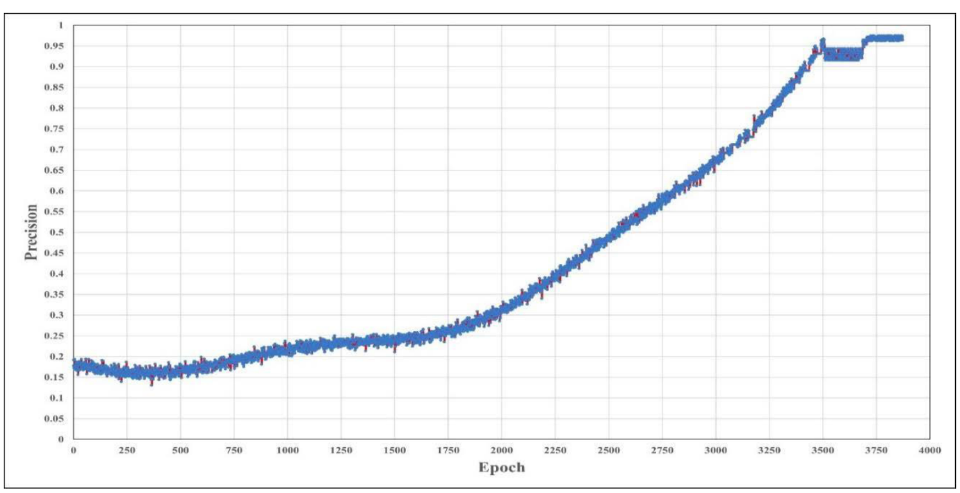

Figure 7—Precision per Epoch

The training process for this algorithm lasted 20 hours, 48 minutes, and 27 seconds on a core i9 12900E CPU with 32GB of RAM. This timescale is particularly valuable, especially due to the efficiency gained from employing the computer's graphics processing unit (GPU) for algorithmic tasks, rather than depending exclusively on the central processing unit (CPU). It is essential to recognize that hardware specifications can significantly impact the length of the training process. Moreover, the data batching procedure, although essential, incurs a delay when information is sequentially relayed to the algorithm's processing unit or kernel, potentially impacting overall processing duration. while contemplating the significant amount of data involved.

An effective method for acquiring deeper insights into the algorithm's performance is through the analysis and interpretation of its output graphs. The generative-adversarial algorithm is one of the most complex and advanced constructs in artificial intelligence. Thus, analyzing the subtleties in its output graphs acts as an essential record, allowing us to gain a more profound comprehension of its complex subcomponents.

To enhance this comprehension, we focus on the validation loss function, which acts as a critical framework for analyzing the algorithm's learning process. Figure 8, meticulously segmented into three distinct portions according to the graph's slope, provides an extensive overview of the algorithm's progression.

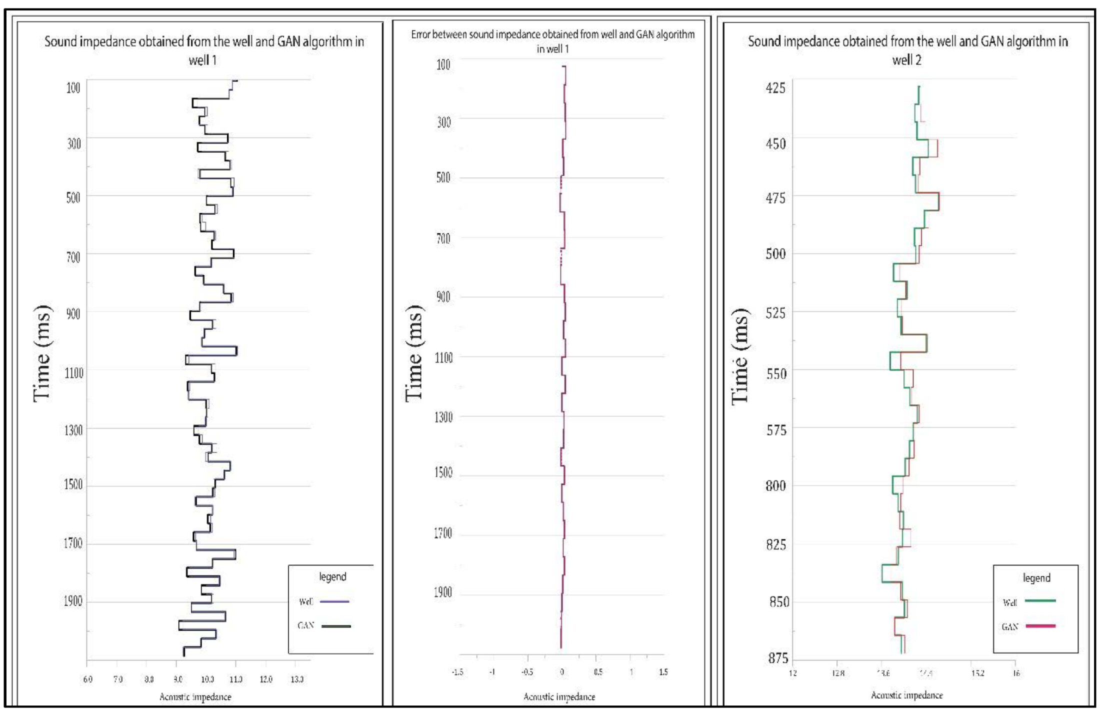

Figure 8—Segmented Validation loss chart

The beginning part of the graph demonstrates a little and gradual alteration in slope. The generative algorithm is currently engaged in its learning process, adeptly creating and producing sound resistance logs. Simultaneously, the adversarial system demonstrates exceptional proficiency in differentiating erroneous data from valid data with amazing precision. In the second phase, we observe the generator algorithm's advancement as it enhances its log-generation abilities. Nevertheless, a significant disruption transpires at the point of transition between part 2 and section 3. This disruption results from the activation of a vital safeguard: the learning protection algorithm. This innovative method mitigates overfitting by periodically deactivating certain neurons inside the algorithm. This proactive strategy avoids the algorithm from being excessively specialized, hence ensuring its adaptability to varied data patterns. Ultimately, in the third part, we witness a harmonious balance between the adversarial and generative algorithms. This delicate equilibrium demonstrates the algorithm's ability to generate precise, high-quality outcomes while preserving data integrity. Essentially, these visual representations of the algorithm's performance, as seen by Figure 8, offer a crucial perspective into its inner workings. operations of this intricate AI system, elucidating its diverse learning dynamics and precisely calibrated self-regulation mechanisms.

As detailed in previous parts, our well dataset consists of 15 wells, 10 of which were essential for the algorithm's training, while the remaining 5 wells were carefully allocated for testing the method. It is crucial to emphasize that these five wells, together with their adjacent seismic track, were not detected by the algorithm. The data collection process deliberately omitted information regarding these particular wells and their corresponding seismic trace. A rigorous approach was implemented to verify the algorithm's results align precisely with the scale of the well data. Initially, the acoustic impedance profiles for each well were meticulously computed. Subsequently, these data were adeptly converted to align with the seismic scale around the well, employing a Backus averaging technique. Subsequent to this essential phase, the aftershocks adjacent to each well were acquired as specified data inputs for the data-adversarial system. The algorithm subsequently produced the inverse, specifically the acoustic impedance, which was meticulously assessed against the acoustic impedance obtained directly from the well logging data. The results of this thorough review have been remarkably gratifying, a reality that will become more apparent as we explore our findings in further depth. A deliberate effort was undertaken to guarantee that the selection of the employed wells was not restricted to a singular geographic area or a narrow temporal span. This strategy decision was motivated by the goal of achieving algorithmic outcomes that cover a wide range, including both high and low levels of acoustic impedance. Figures 9 and 10 illustrate the acoustic impedance estimations alongside the sound resistance obtained from well logging. Furthermore, each estimation is accompanied with its corresponding error rate. The mean squared error (MSE) figures indicate the precision of the algorithm's predictions. For well number one, the MSE is 0.10615 acoustic impedance units, but for well number two, it is 0.10819 acoustic impedance units. This trend persists, with the MSE values for each well demonstrating the algorithm's efficacy in estimating acoustic impedance throughout the sample. Specifically, well number thirty-five exhibits a mean squared error of 0.10169 acoustic impedance units, well number thirty-seven attains 0.12529 acoustic impedance units, and well number fifty records 0.11555 acoustic impedance units.

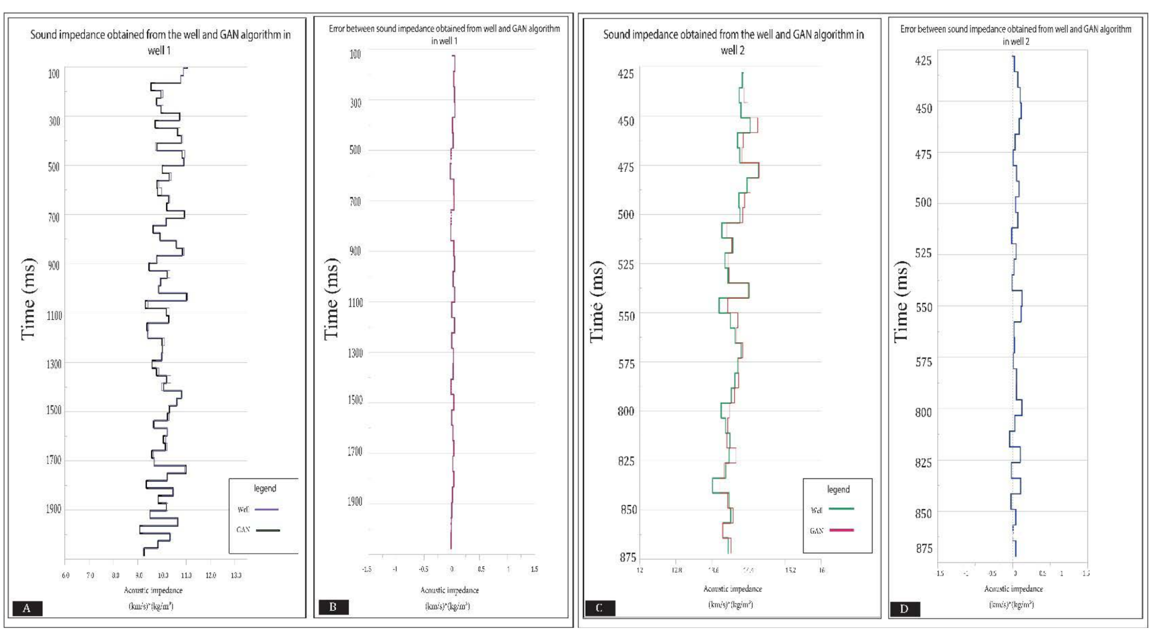

Figure 9—Comparing the impedance obtained from the well and the GAN algorithm along with its error. (A&B) Well No.1; (C&D) Well No.2

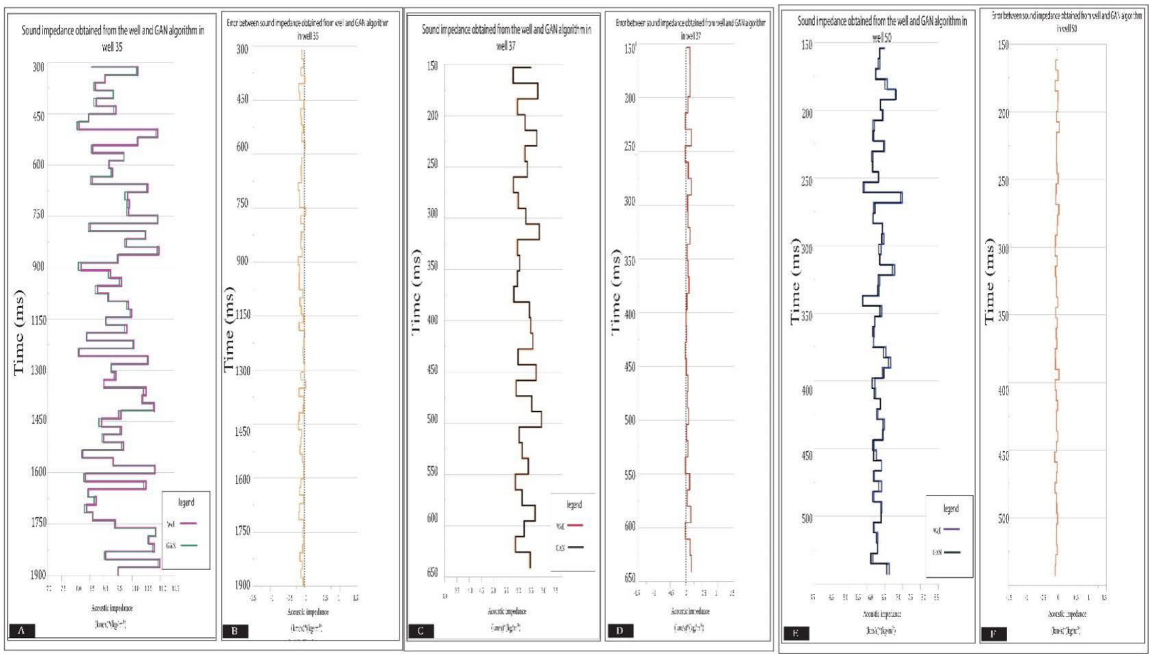

Figure 10—Comparing the impedance obtained from the well and the GAN algorithm along with its error. (A&B) Well No.35; (C&D) Well No.37; (E&F) Well No.50

These results highlight the algorithm's exceptional ability to estimate acoustic impedance values, illustrating its strength and dependability across various well data contexts.

Figures 11 and 12 detail the findings of seismic cube inversion.

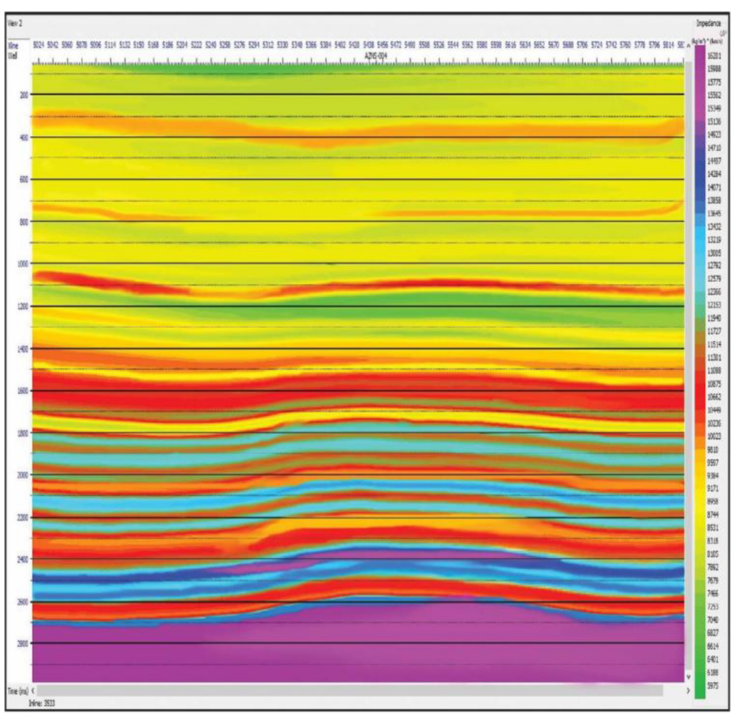

Figure 11—In-line number 3533-GAN Inversion results

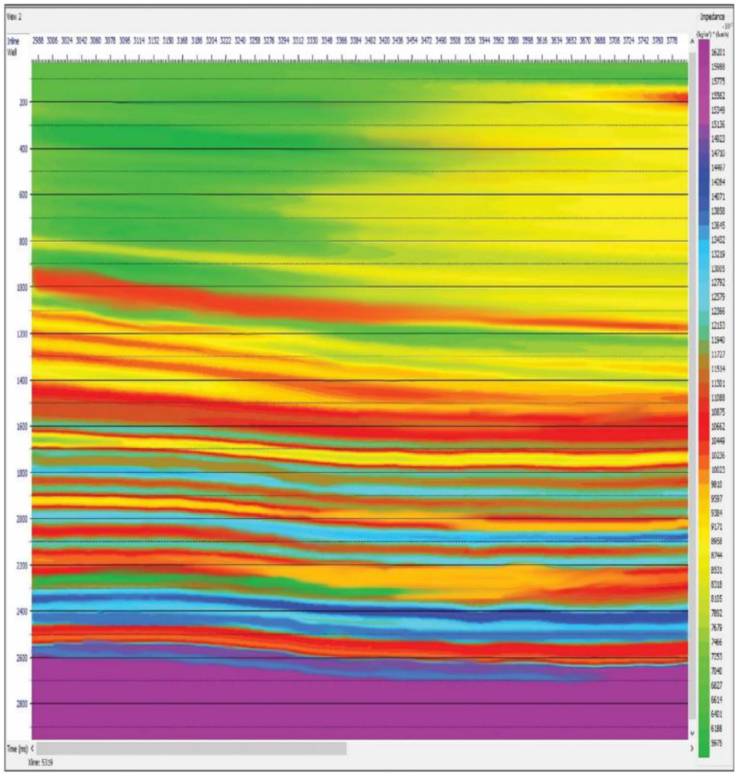

Figure 12—X-line number 5319 – GAN Inversion results

In seismic inversion, the generative-adversarial algorithm differs dramatically from existing approaches in several ways. A crucial difference is the absence of wavelet extraction, a key indicator of its uniqueness. Using wavelet extraction in traditional methods may reduce accuracy. Wavelet preparation requires manual intervention, with operators iterating to find the best wavelet. Traditional seismic inversion results reflect human ability and precision.

Furthermore, seismic and well-logging operations face ongoing challenges due to noise. Many older procedures are prone to noise interference, compromising result accuracy. Increased noise might make a dataset unreliable, requiring repeated data capture. However, the generative-adversarial deep learning approach eliminates the necessity for wavelet extraction during inversion. The inversion procedure involves generating a log at each trace point and determining the acoustic impedance log. Since the generative-adversarial system may learn, external wavelet extraction is not necessary. After training, it can differentiate noise from the original data, considerably reducing its impact. The program learns to recognize data interactions and reduces noise during production through memory during training. The generative-adversarial algorithm inversion approach offers a more efficient workflow by eliminating secondary operator calculations, unlike the color inversion method. In the next sections, we will compare the generative adversarial algorithm's inversion results to band-limited approaches and model-based inversion to demonstrate its unique characteristics (Figures 13 and 14).

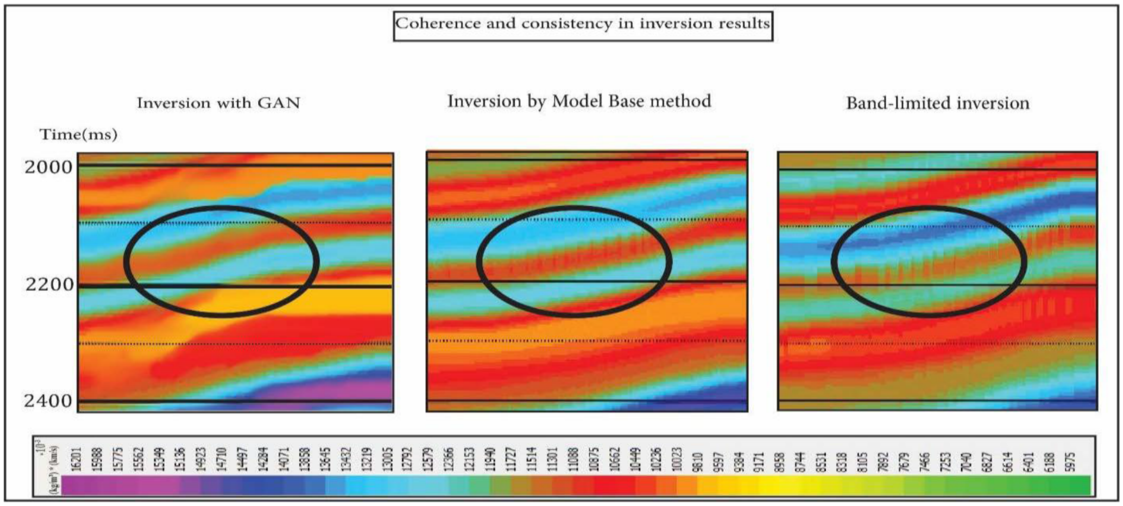

Figure 13—Comparing the consistency of seismic inversion results using three GAN

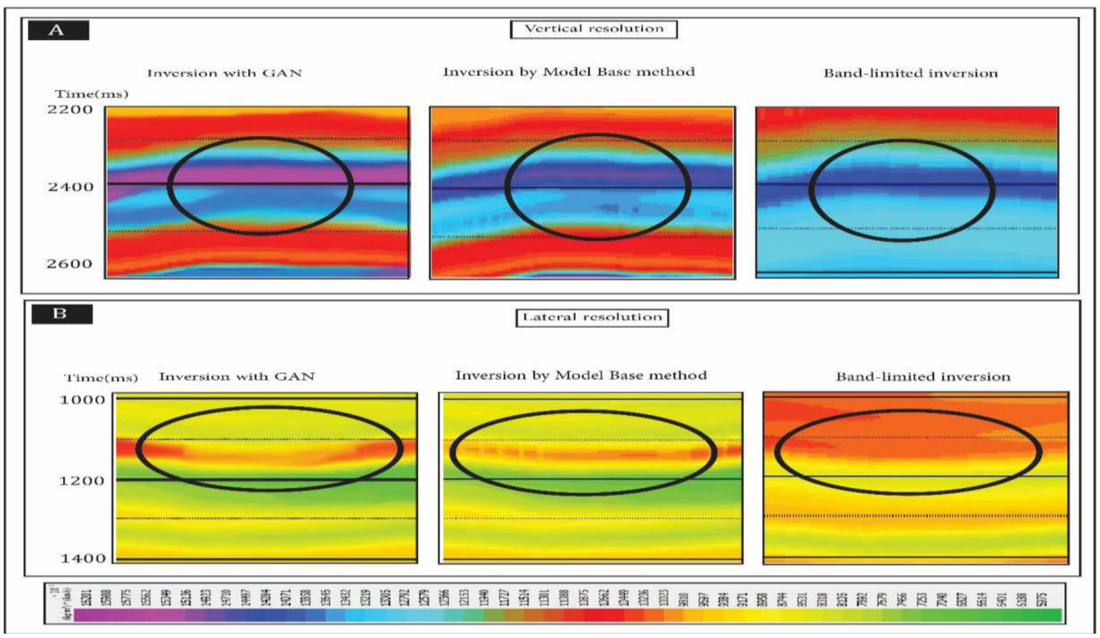

Figure 14—Compare resolution of seismic inversion results using three GAN algorithms methods, band limited method and model base method. (A) Vertical resolution; (B) lateral resolution

We examined the correlation coefficients, a key indicator of alignment, between generative-adversarial algorithm-generated datasets and well-extracted acoustic impedance data. Additionally, we critically analyze results from band-limited and model-base techniques (Table1).

Table 1—correlation coefficient between the results of three methods of inversion with well data

Methods of inversion	GAN	Model-base	Band -limited
Well No.2	0.96143	0.78652	0.66614
Well No.35	0.98994	0.83537	0.71285

Results are presented in Figures 15–and Figures 16. Note that the two wells under study are a critical part of the test dataset. It is noteworthy that the generative-adversarial algorithm was not trained on this specific data set. The algorithm's capacity to generalize to new data is shown by this critical distinction.

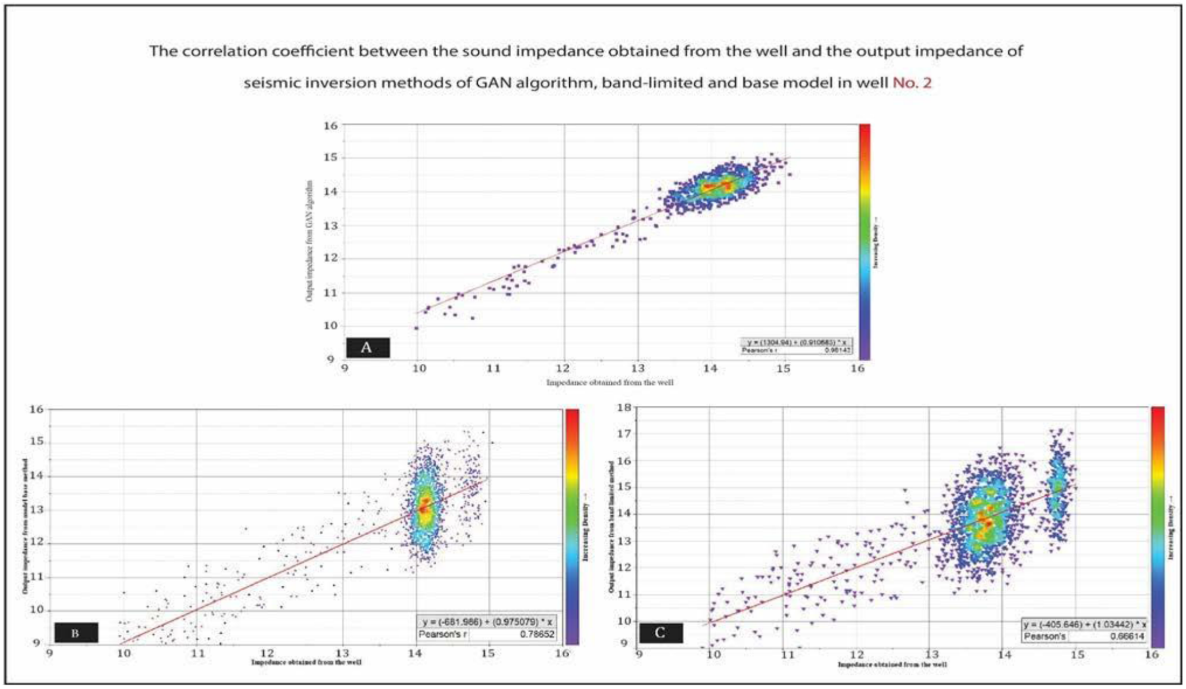

Figure 15—Correlation coefficient between the acoustic impedance obtained from well number2 and the output impedance of seismic inversion methods of generator-adversarial algorithm, band limited and model-based. (A) GAN(B) Model-based method; (C) Band-Limited Method

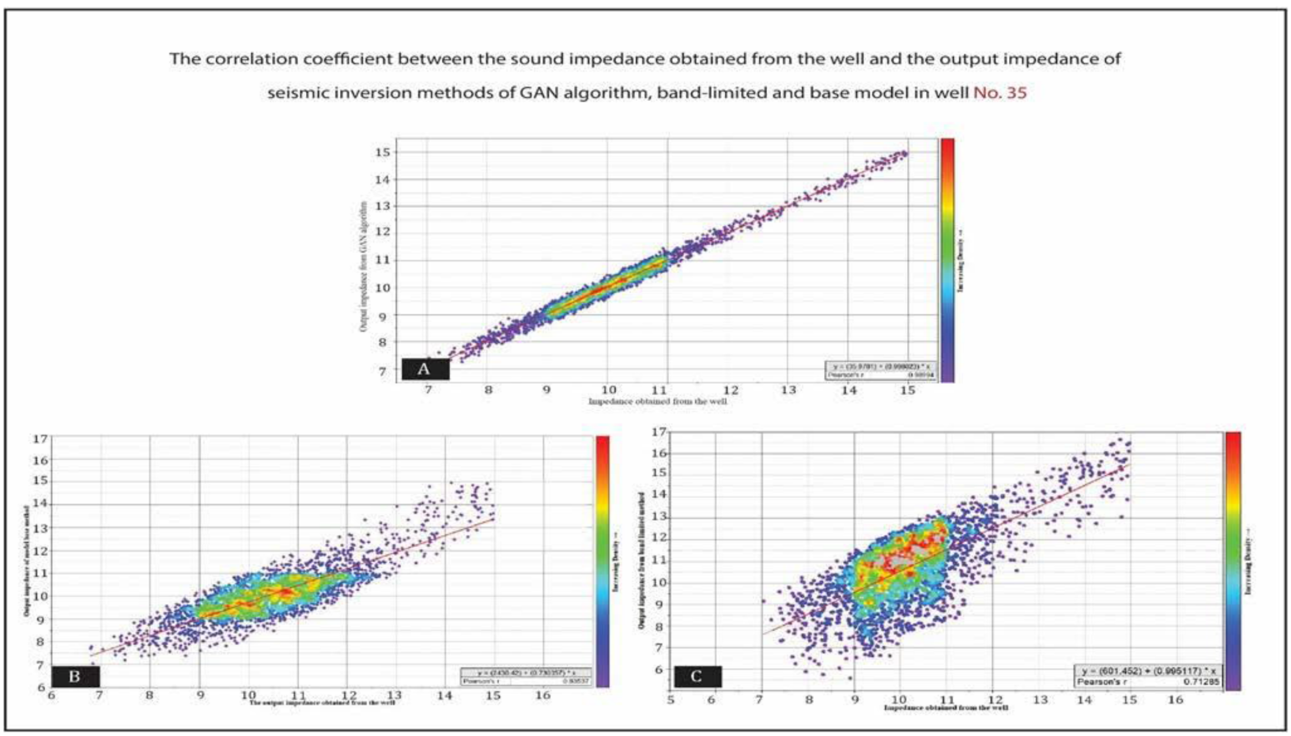

Figure 16—Correlation coefficient between the acoustic impedance obtained from well number2 and the output impedance of seismic inversion methods of generator-adversarial algorithm, band limited and model-based. (A) GAN(B) Model-based method; (C) Band-Limited Method

Conclusion

Seismic inversion is an efficient tool for investigating hydrocarbon deposits, and ongoing improvement and the creation of novel approaches are crucial for enhancing productivity and reducing risks in exploratory operations. In this context of advancement, the Generative Adversarial Deep Learning algorithm has arisen as a state-of-the-art and exceptionally effective instrument for producing highly realistic models. This approach improves precision and enhances computational efficiency when utilized for seismic data inversion. Furthermore, its advancement within the Python programming ecosystem guarantees global accessibility for specialists, promoting an open-source culture for continued enhancement and evolution. Notwithstanding its intricate architecture, the Generative Adversarial technique is accessible, with the training and execution stages delineated, rendering seismic cube inversions as uncomplicated as a few mere clicks.

The Generative Adversarial Deep Learning algorithm is an effective tool for producing highly realistic models and markedly decreasing errors in acoustic impedance measurements relative to traditional methods like Band-limited and Model-based inversion, closely aligning with impedance values obtained from well data. It possesses remarkably high vertical and lateral resolution capabilities for calculating acoustic impedances, facilitating the identification of subsurface layers based on impedance variations.

Moreover, its structural integrity renders it impervious to noise, so averting mistake propagation resulting from noise in the outcomes. The GAN algorithm demonstrates a significant connection between its output impedance and actual values, obtaining a maximum Mean Squared Error (MSE) of 0.12529 acoustic impedance units, in contrast to 5.6773 observed in alternative inversion approaches, resulting in an accuracy of 97.5%.

Future research must prioritize the optimization of temporary memory management to avert bottlenecks and improve algorithm training efficiency. Investigating more efficient alternatives to the Backus method may diminish code complexity and enhance accuracy. Comparative analyses with alternative deep learning and machine learning methodologies can enhance the algorithm and underscore its advantages. Exploring new sub-network topologies beyond fully connected networks demonstrates potential, perhaps uncovering innovative insights and enhancing performance.

In summary, the Generative Adversarial Deep Learning method signifies a promising advancement in seismic inversion and subsurface exploration. Its capacity for enhancement and advancement, along with its durability and efficacy, renders it an invaluable instrument for both present and future specialists in the domain. Continued research and innovation in this field are essential for uncovering enhanced insights and capabilities in hydrocarbon reserve exploration.

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